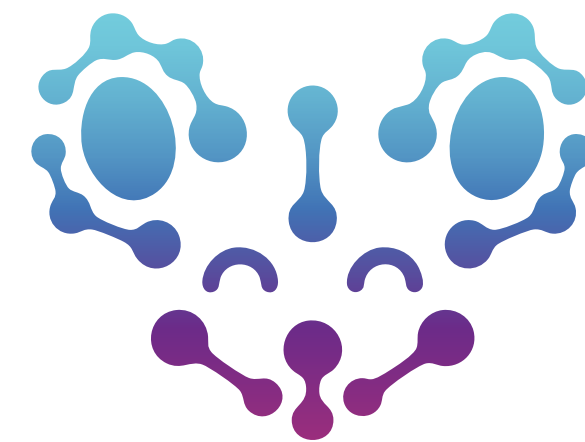


Measuring Progress in Neuro Foundation Models with BrainWideBench

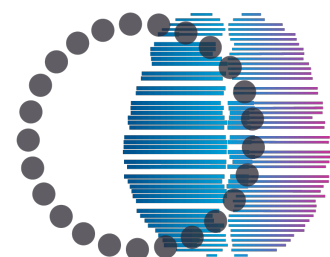


Alexandre Andre & Shivashriganesh Mahato

NerDS Lab, University of Pennsylvania

September 24, 2026

ARNI Annual Retreat 2026



INTERNATIONAL
BRAIN
LABORATORY



Toward Broader Neural Models



Toward Broader Neural Models



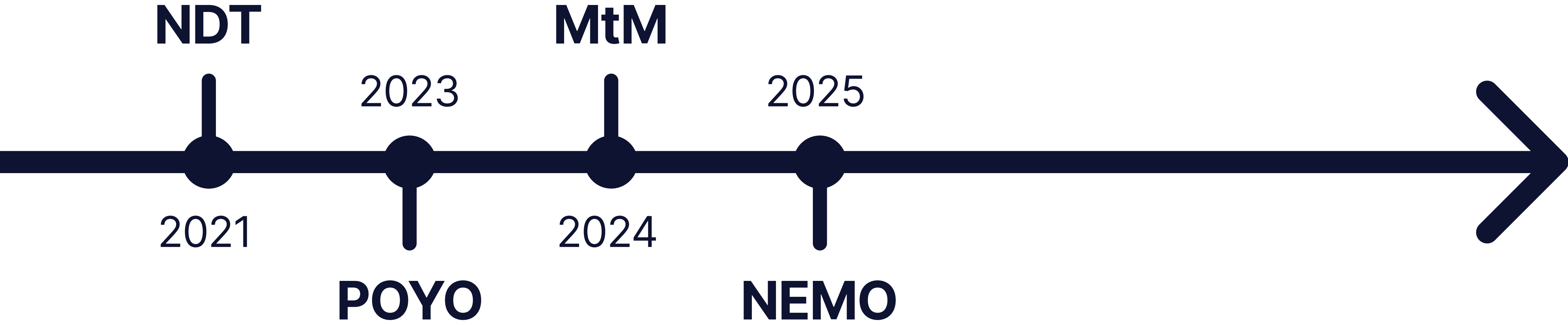
Toward Broader Neural Models



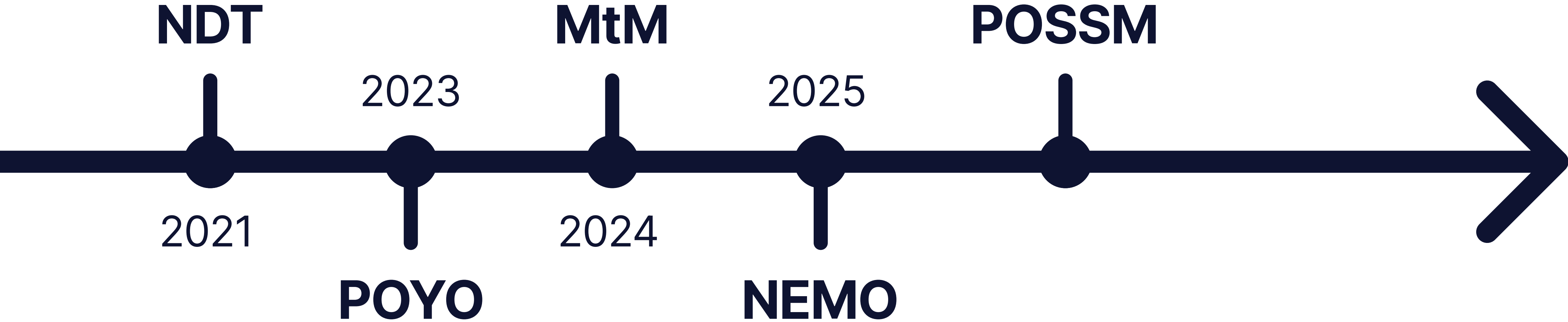
Toward Broader Neural Models



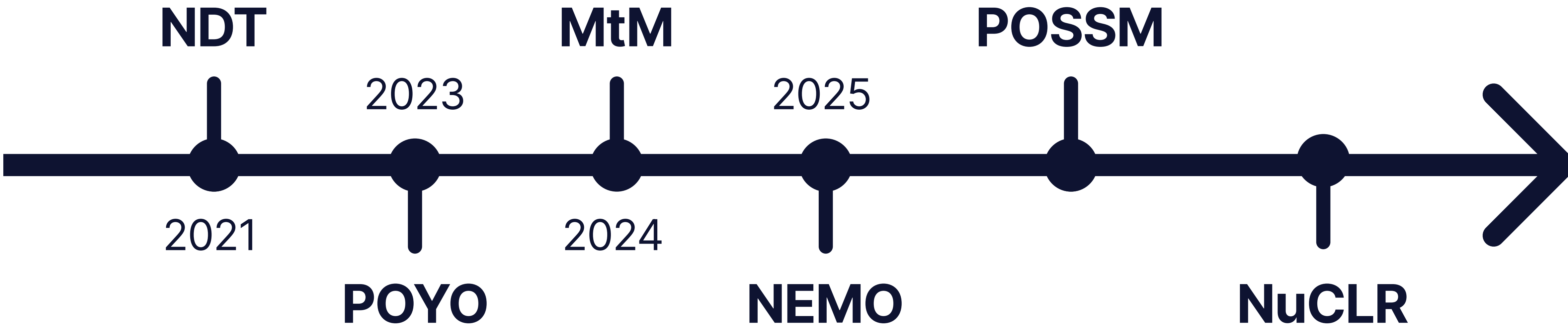
Toward Broader Neural Models



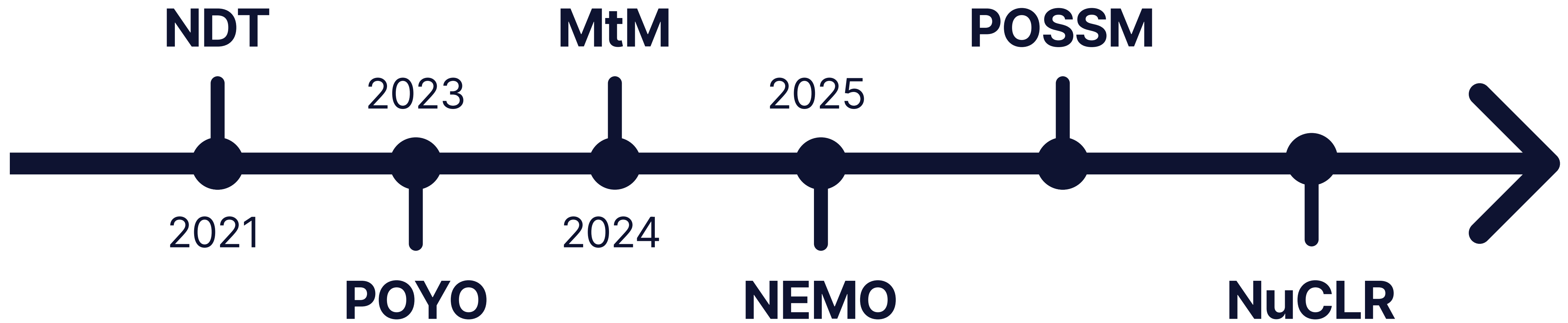
Toward Broader Neural Models



Toward Broader Neural Models

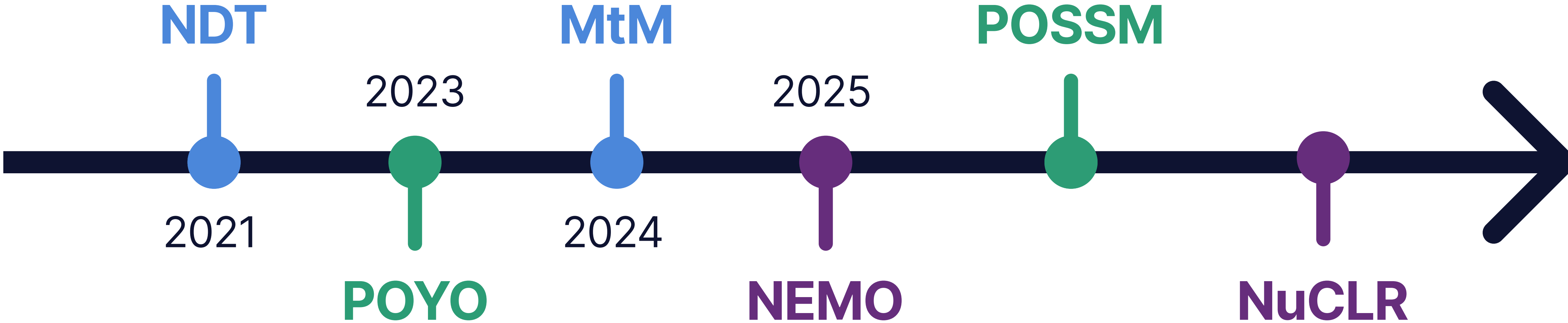


Toward Broader Neural Models



Diverse goals: decoding, forecasting, cell-type & brain region classification, real-time inference

Models Capture Different Aspects





Dynamics



Behavior



Anatomy

A General Model Should Do All Three



Dynamics



Behavior



Anatomy



Measuring Progress

Problem: no shared evaluation

Measuring Progress

Problem: no shared evaluation

- **Costly:** each new model re-implements baselines and preprocessing

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- **Standardized splits:** held-out animals/sessions + fixed train/val/test

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Goal: a common benchmark

- **Standardized splits:** held-out animals/sessions + fixed train/val/test
- **Unified evaluation:** generalization across animals and tasks
- **Strong baselines:** well-tuned single-session and pretrained models
- **Open & easy to use:** plug in a new model with minimal effort

BrainWideBench

Dataset

139 mice
452 sessions

276 brain regions
830k+ unit-hours

Baseline

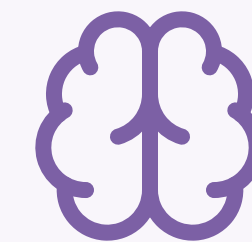
18 single-session
6 pretrained (**4**/6 are from ARNI)



Dynamics



Behavior

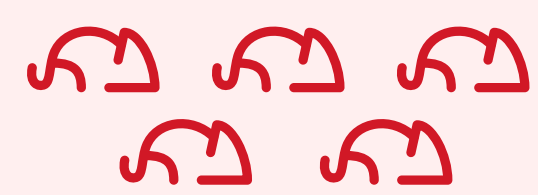


Anatomy

Common Evaluation Framework

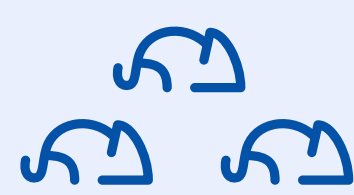
Pretrain

(126 Mice · 423 Sessions)



Eval

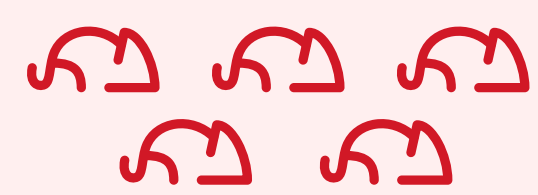
(13 Mice · 29 Sessions)



Common Evaluation Framework

Pretrain

(126 Mice · 423 Sessions)



Model

···· Optional



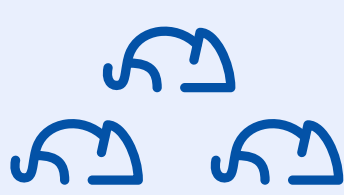
Input



Target

Eval

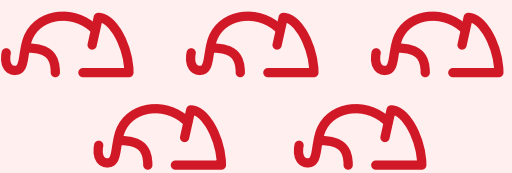
(13 Mice · 29 Sessions)



Common Evaluation Framework

Pretrain

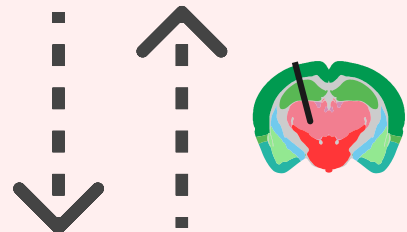
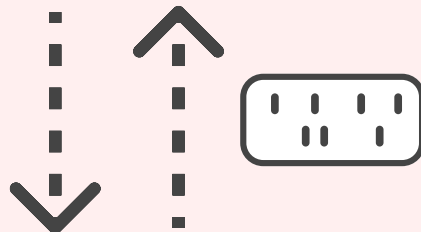
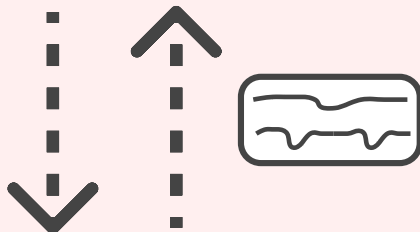
(126 Mice · 423 Sessions)



Behavior

Spikes

Anatomy

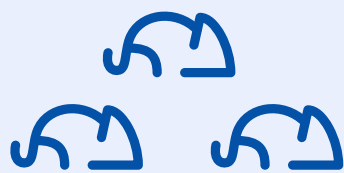


Model

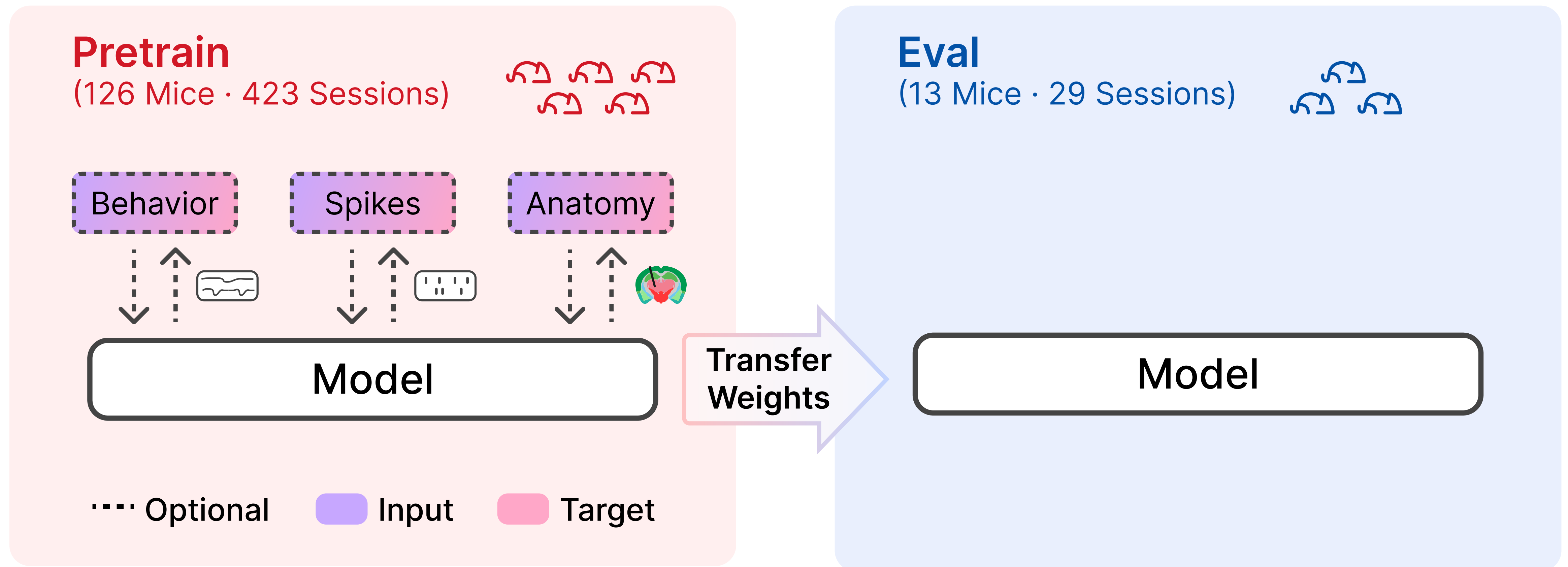
···· Optional  Input  Target

Eval

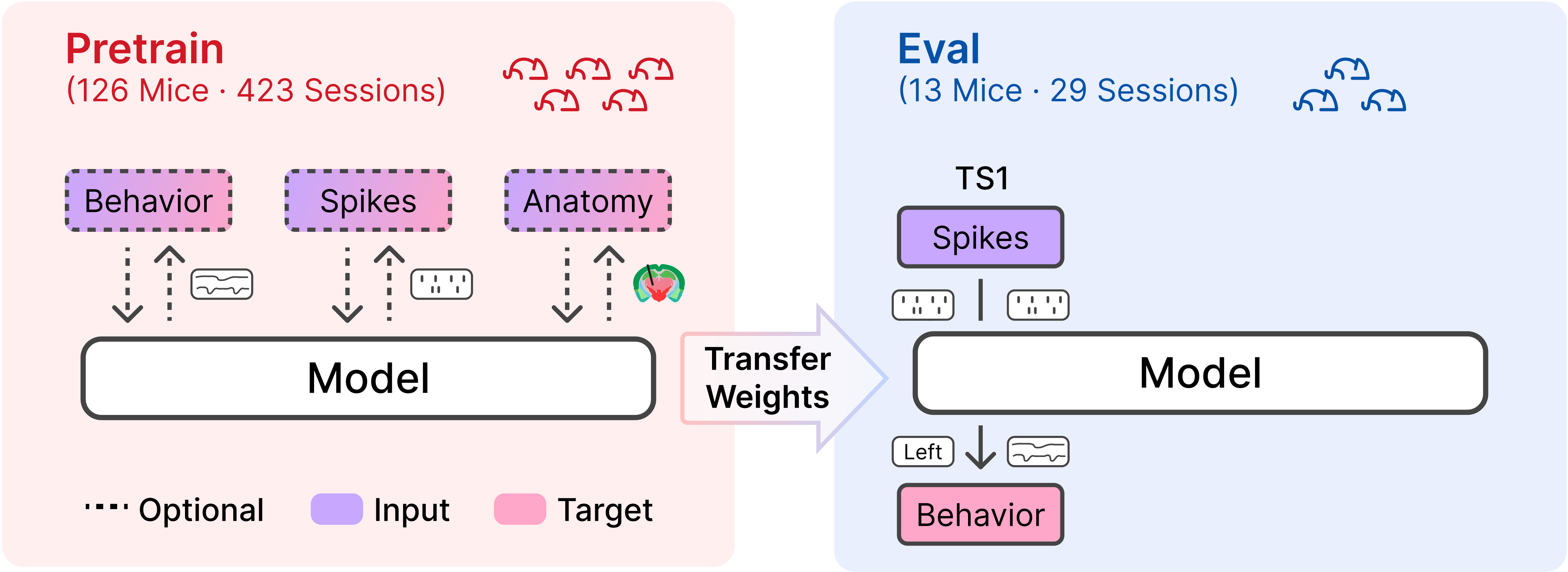
(13 Mice · 29 Sessions)



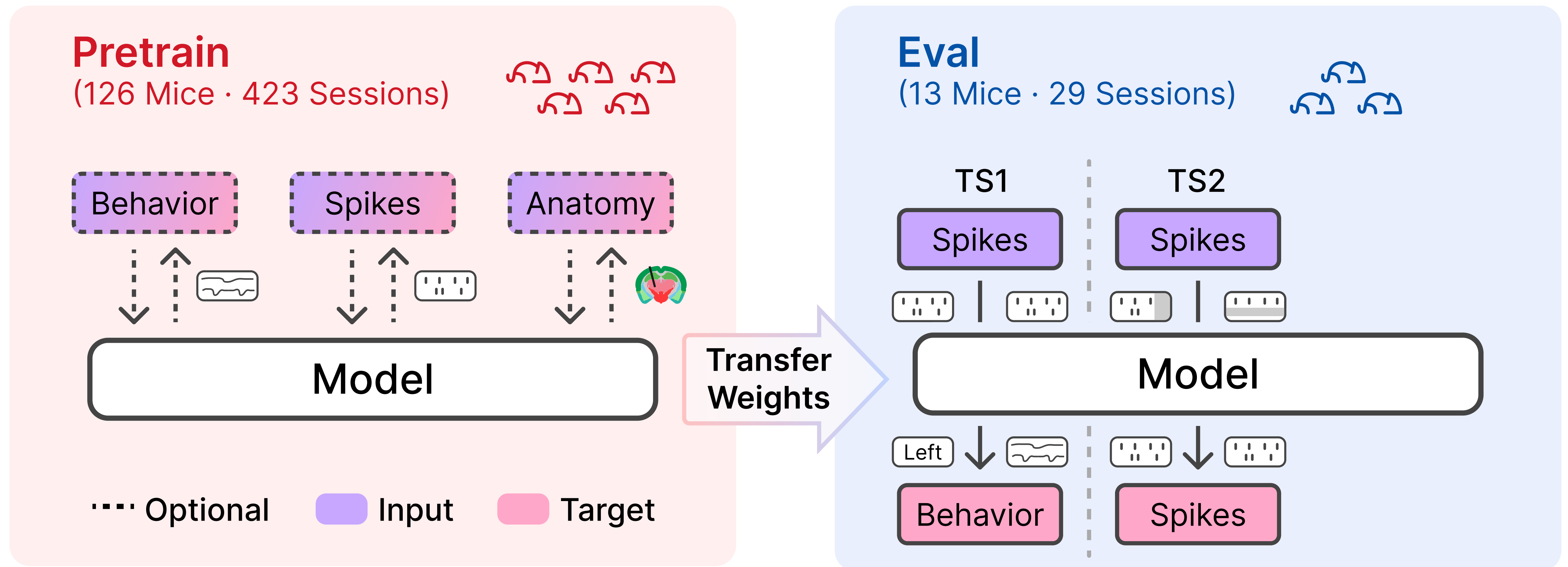
Common Evaluation Framework



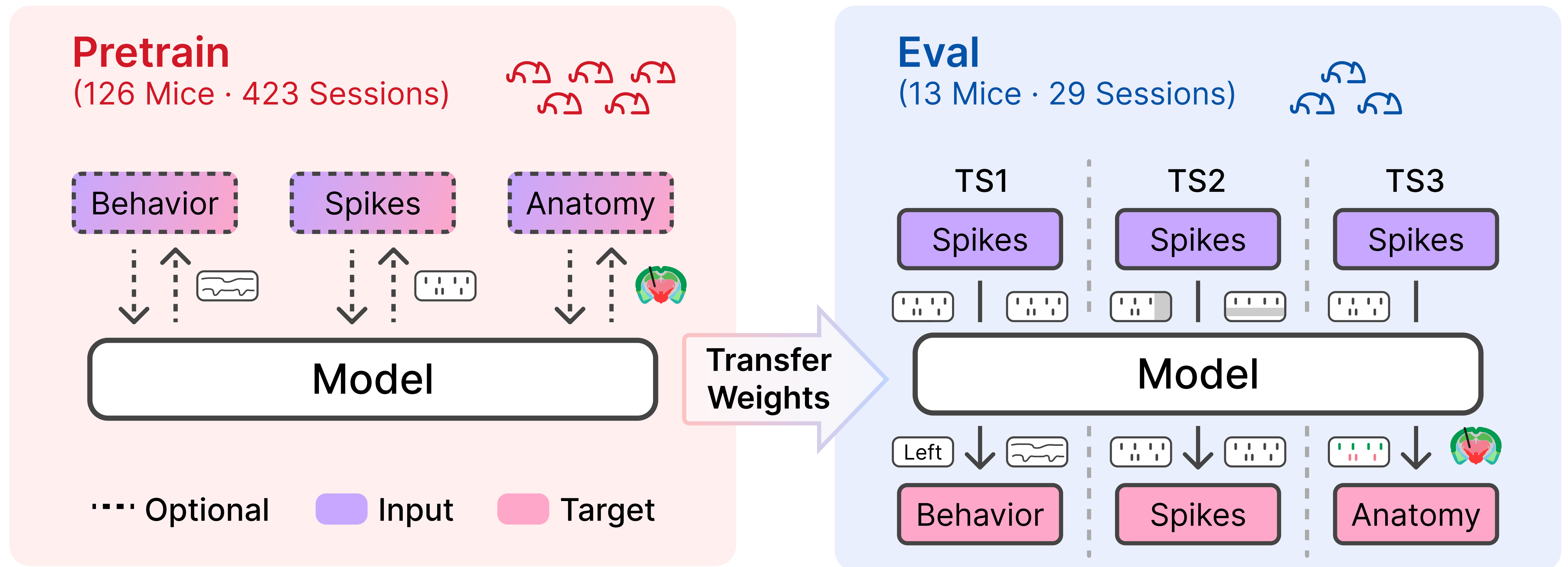
Common Evaluation Framework



Common Evaluation Framework



Common Evaluation Framework



Pretrained Baselines Within the Benchmark Framework

 Model Input  Model Target

NDT



POYO · POSSM



MtM



NuCLR · NEMO



Pretrained Baselines Within the Benchmark Framework

 Model Input  Model Target

NDT



POYO · POSSM



MtM



NuCLR · NEMO



Pretrained Baselines Within the Benchmark Framework

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NDT



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MtM



NuCLR · NEMO



Pretrained Baselines Within the Benchmark Framework

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Pretrained Baselines Within the Benchmark Framework

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POYO · POSSM



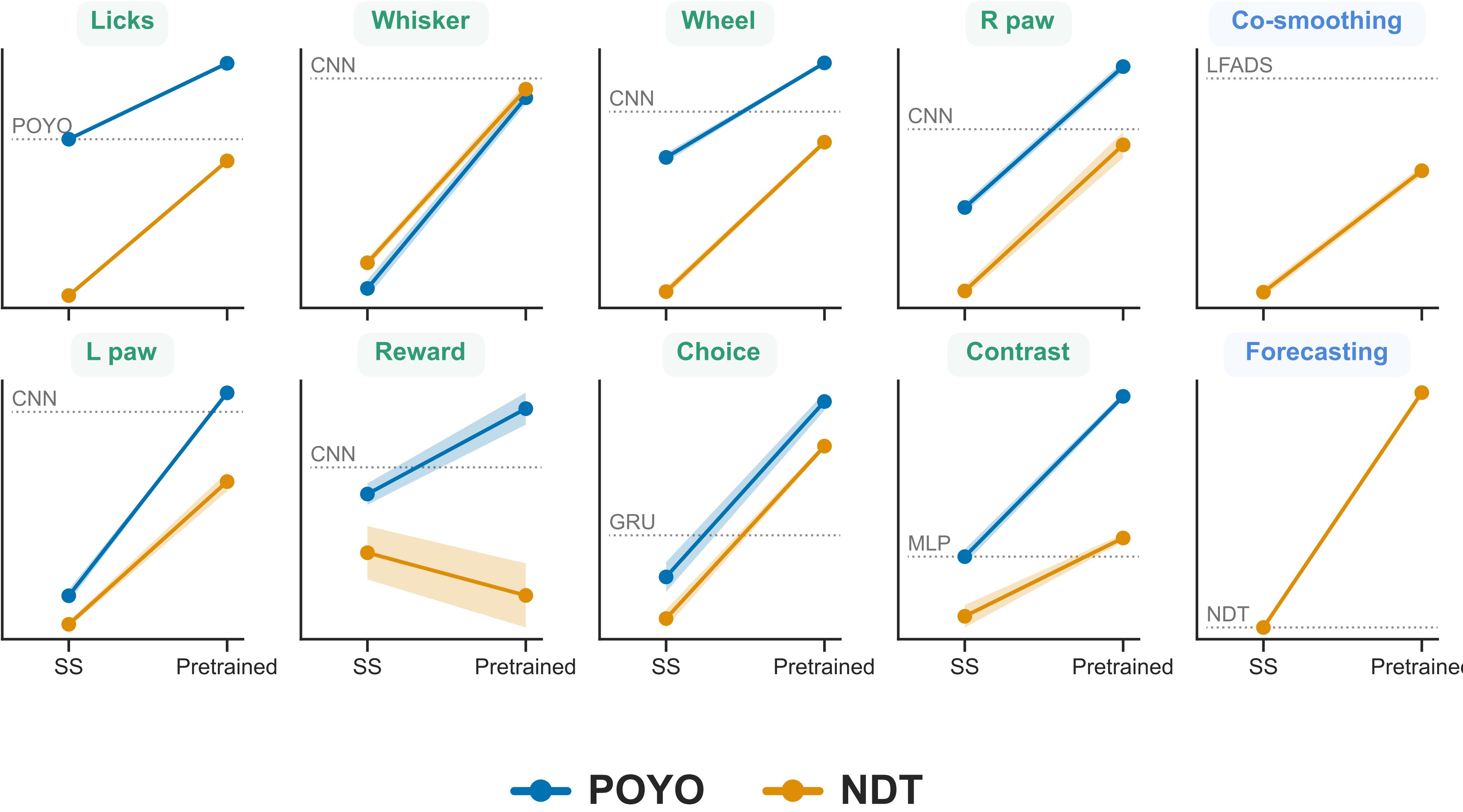
MtM



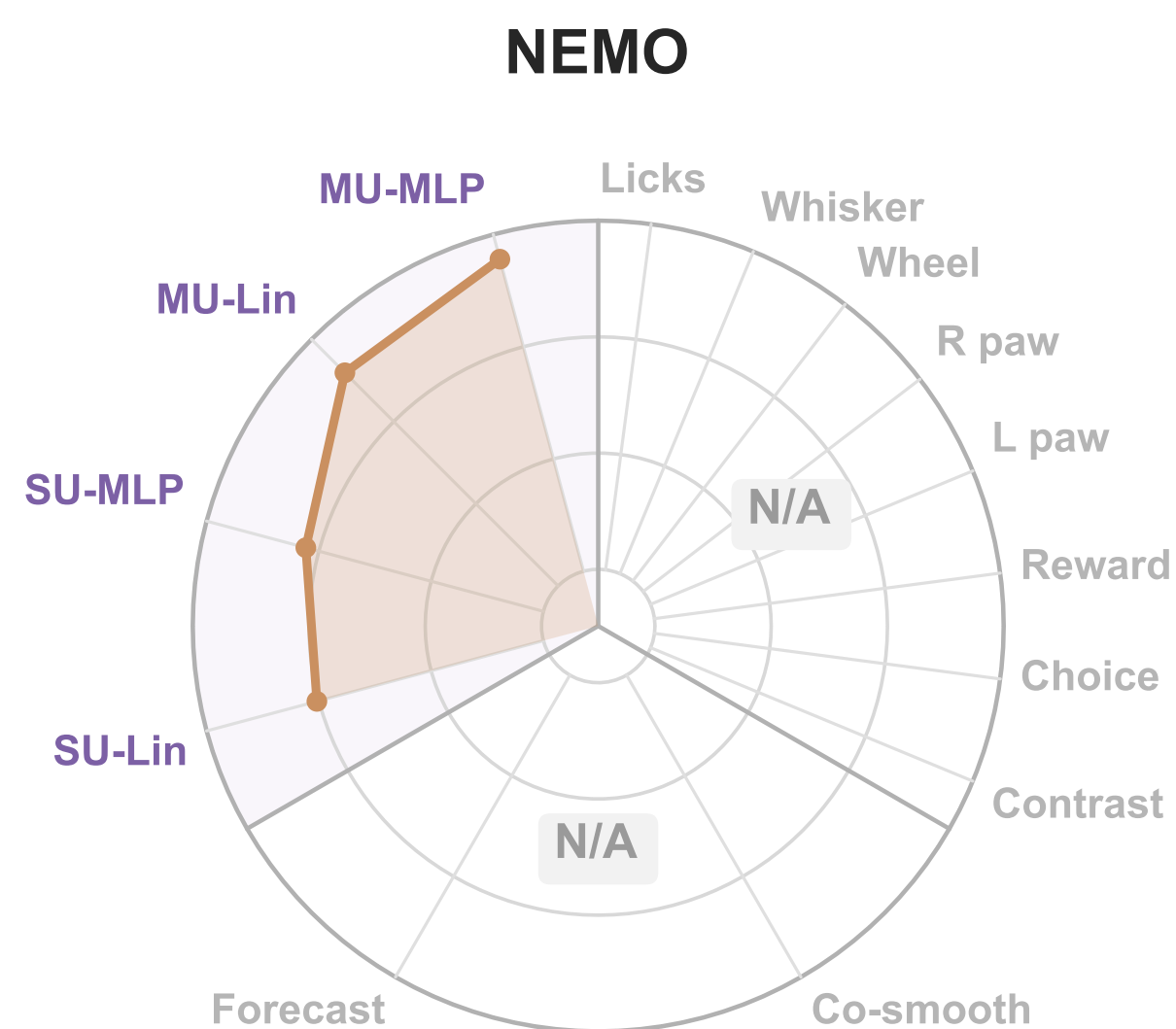
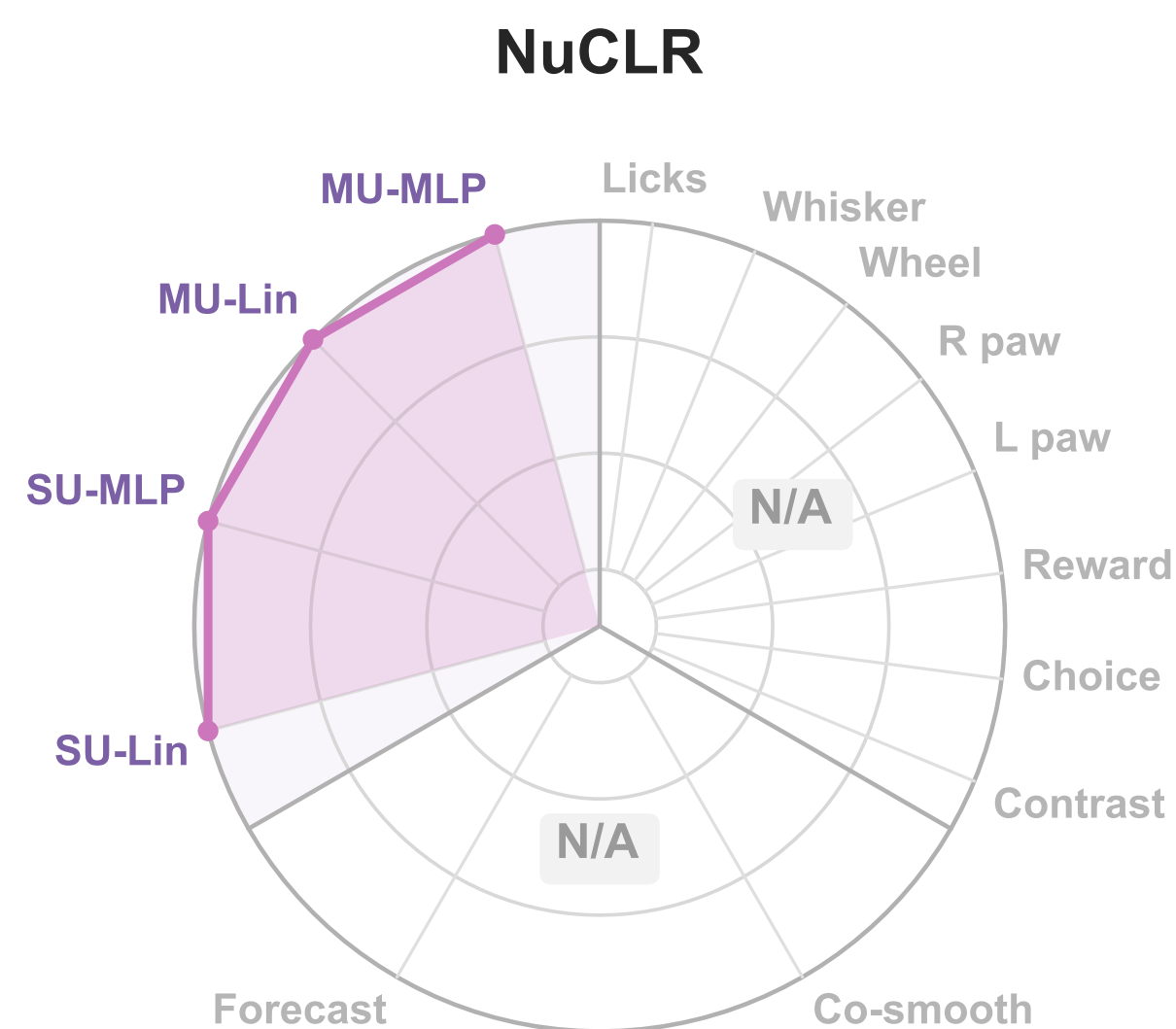
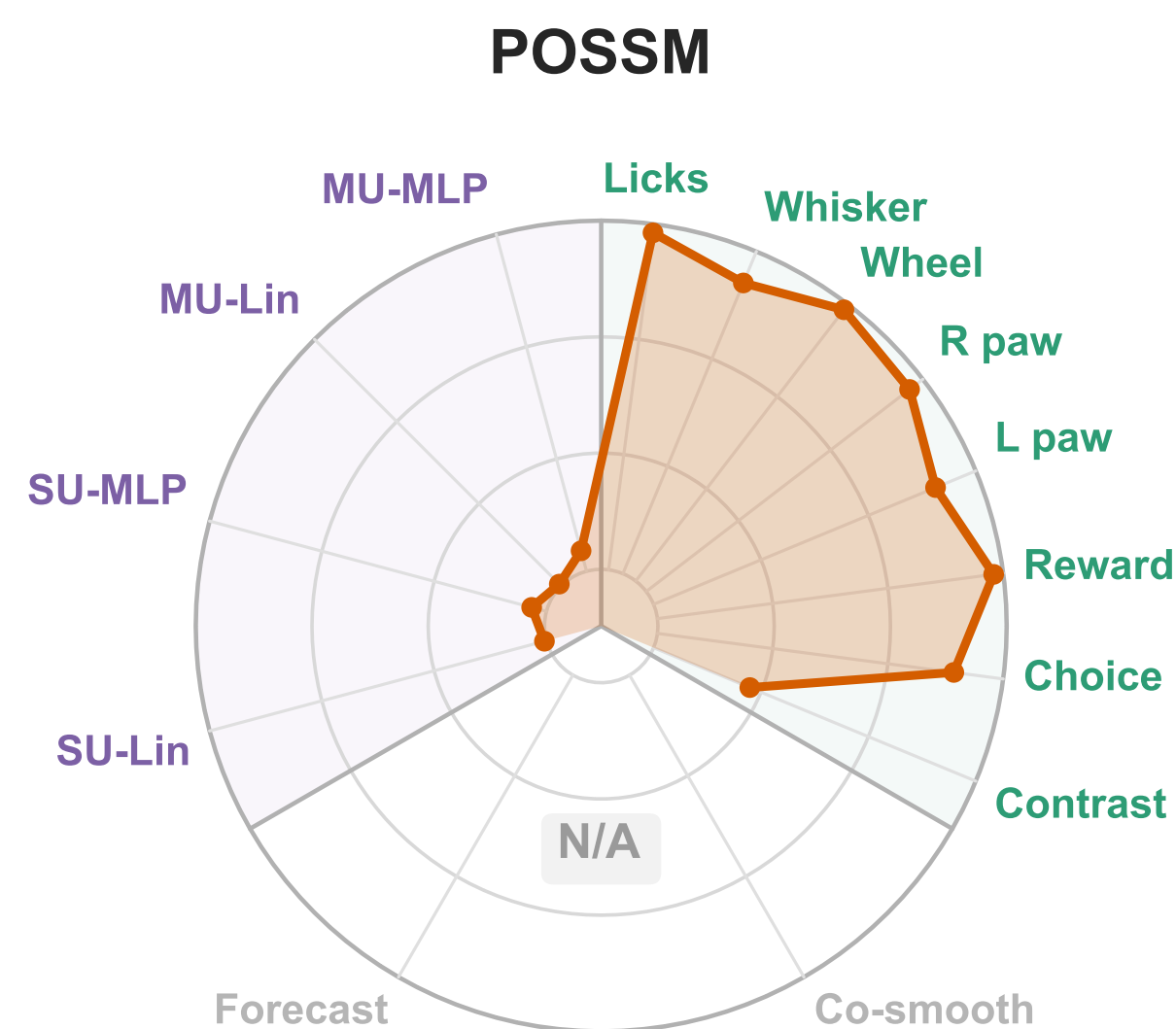
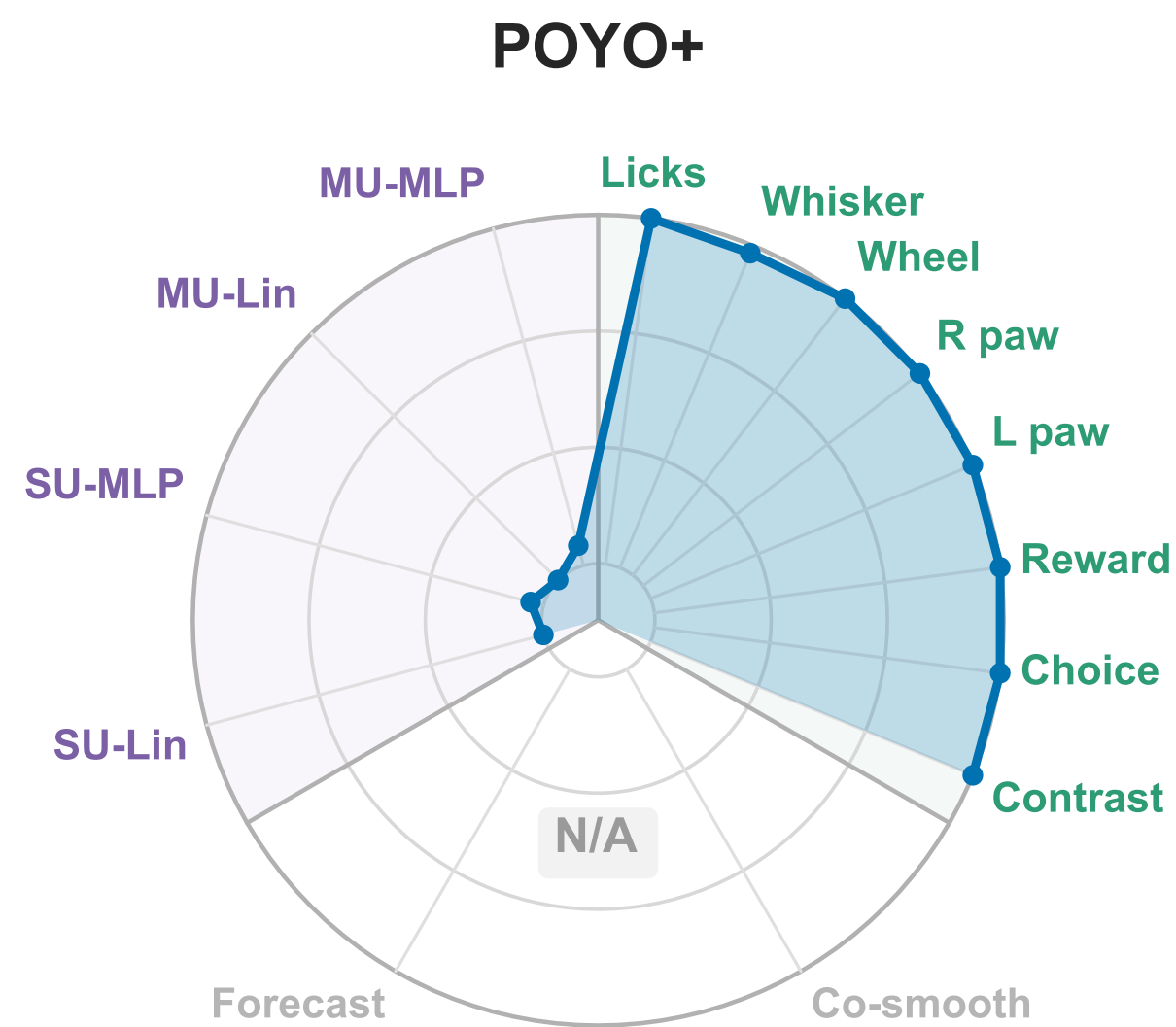
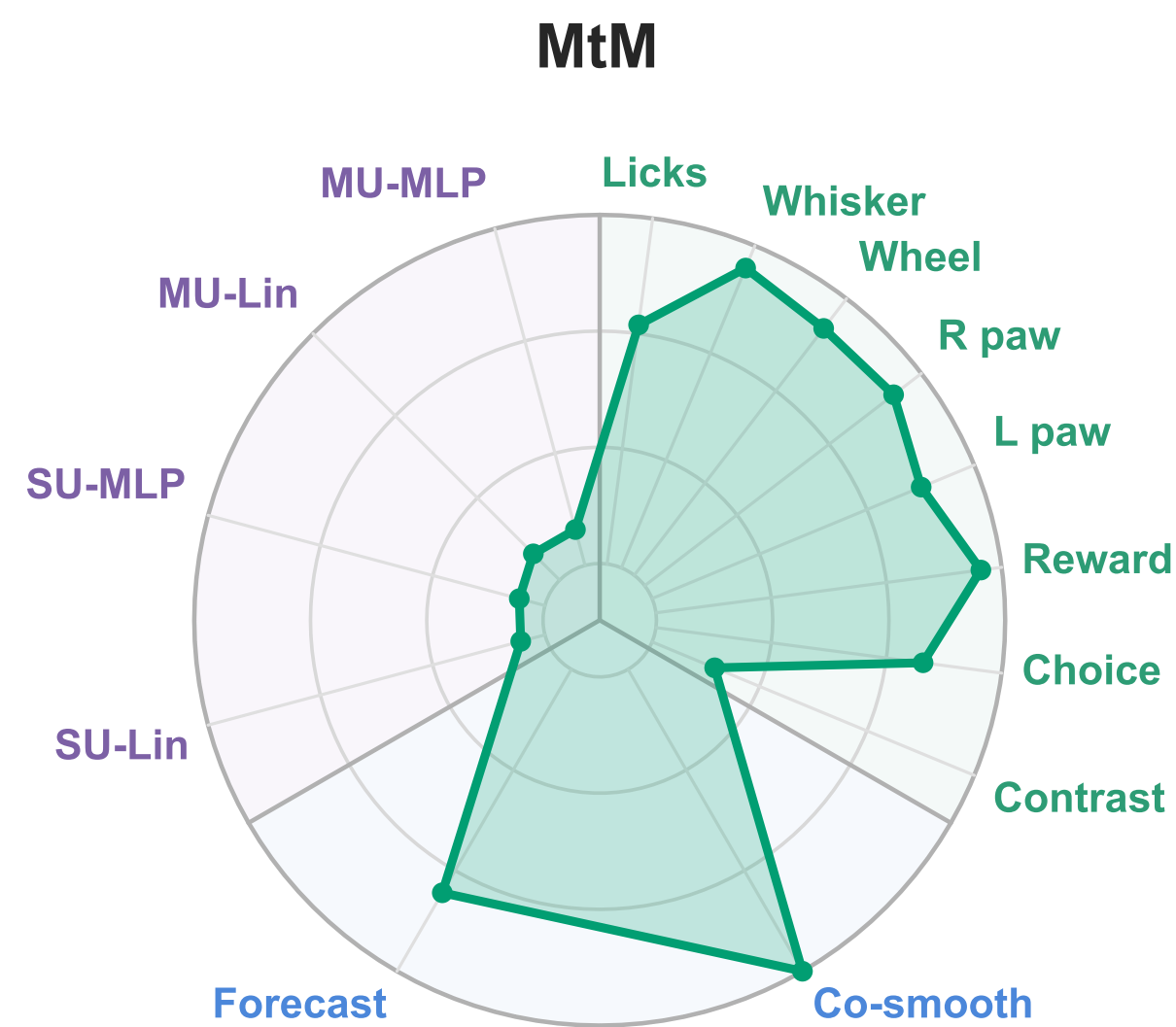
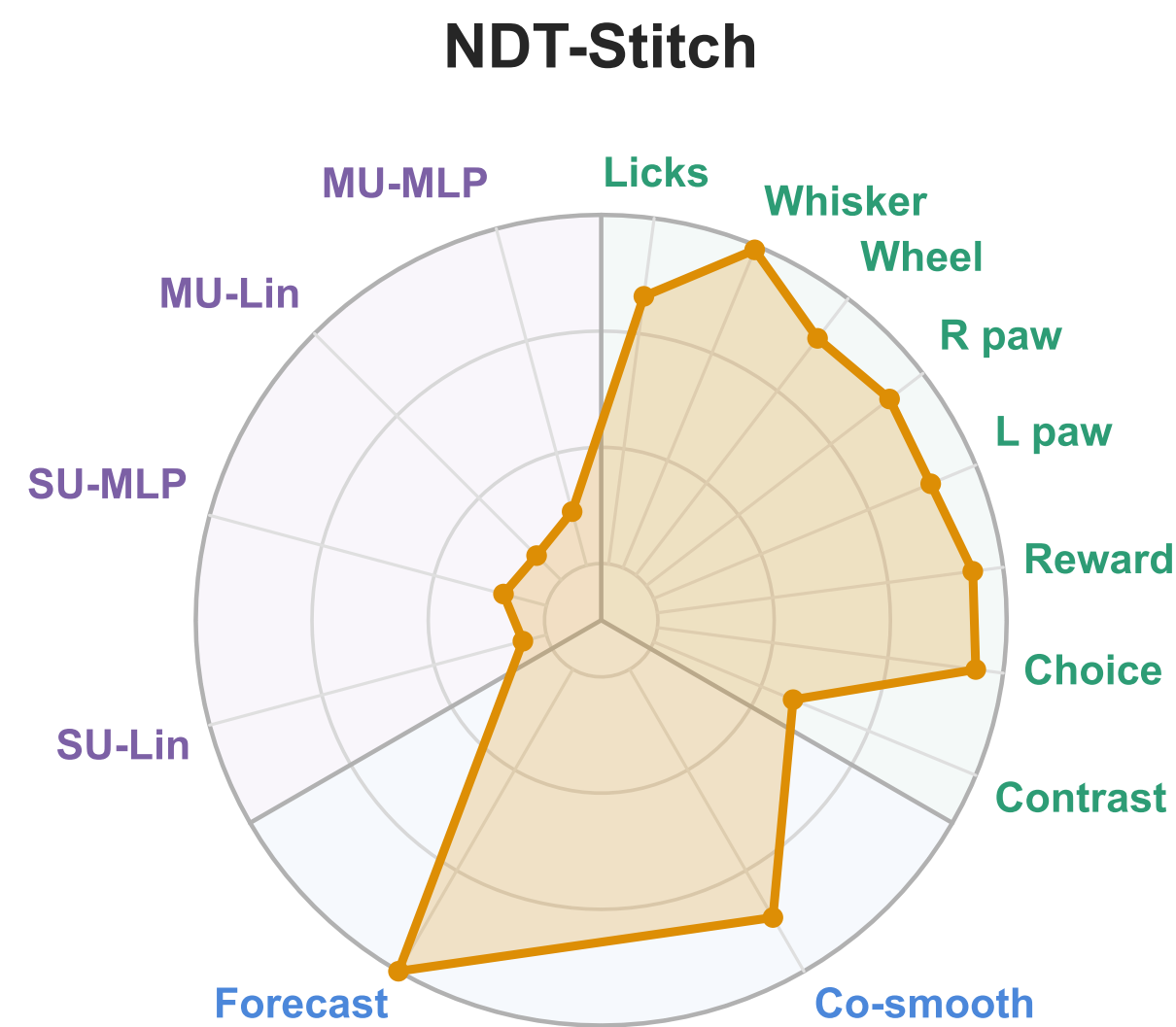
NuCLR · NEMO



Key Results: Pretraining Helps Across Tasks



Key Results: No free lunch (yet!)



Ways to participate

Less time processing data and building evaluation, more time on modeling ideas



Get started

[brainbench-org.github.io/
ibl-bwb-project/](https://brainbench-org.github.io/ibl-bwb-project/)

Leaderboard

Public, completed submissions scored against the benchmark's evaluation set.

Suites ▾

Select the suites or a combination of tasks to include in the ranking.

TS1 TS2 TS3 Add task

TS1 Choice × TS1 Left paw × TS1 Licking × TS1 Reward × TS1 Right paw × TS1 Stimulus ×
TS1 Wheel × TS1 Whisker × TS2 Co-smooth × TS2 Forecast × TS3 Cosmos ×

Filters ▾

Apply filters to restrict models or tasks included in the ranking. Click more filters to show all

Is pretrained? Pretrained in modalities Pretrained out modalities + More filters

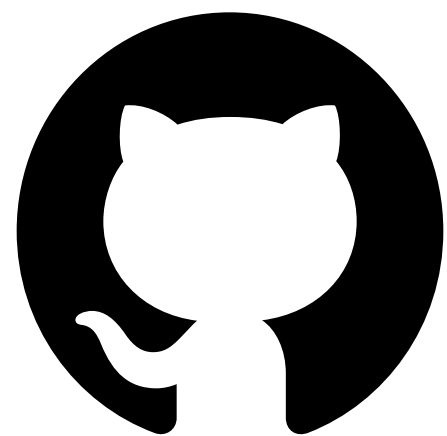
2 models ranked across 11 tasks, 20 unranked

Click Compare models and select up to 6 models to compare their performance. Compare models

#	Model	Choice BAcc	Left paw R ²	Licking Poisson D ²	Reward BAcc	Right paw R ²	Stimulus BAcc	Wheel R ²	Whisker R ²	Co- smooth Poisson D ²	Forecast Poisson D ²	Cosmos Macro F1
1	NDT-Stitch Pretrained Brain Wide Bench	0.674 ± 0.001	0.172 ± 0.003	0.525 ± 0.002	0.863 ± 0.006	0.219 ± 0.005	0.229 ± 0.001	0.310 ± 0.002	0.384 ± 0.003	0.157 ± 0.001	0.177 ± 0.000	0.187 ± 0.008
2	MtM Pretrained Brain Wide Bench	0.646 ± 0.004	0.167 ± 0.003	0.469 ± 0.004	0.874 ± 0.003	0.224 ± 0.003	0.213 ± 0.002	0.323 ± 0.006	0.362 ± 0.004	0.191 ± 0.001	0.131 ± 0.001	0.159 ± 0.009

Submit your model

Open Science: Path to General Neural Models



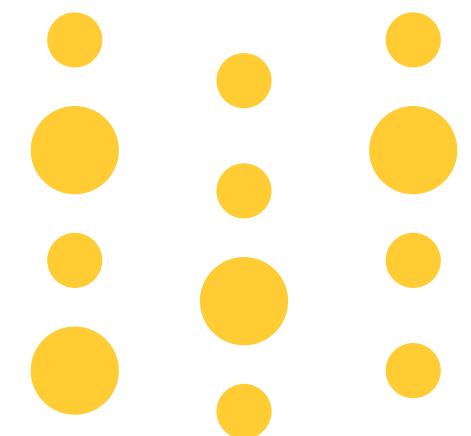
github.com/brainbench-org/ibl-bwb



torchbrain.org



hf.co/collections/nerdslab/ibl-bwb



wandb.ai/ibl-bwb/projects

Questions?

